Website Optimization Report

Site: https://www.edu123.in

Prepared For: Web Development Internship

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1. Page Speed & Performance Overview

Tools used: Google PageSpeed Insights, Chrome DevTools, Lighthouse Report

Metric	Desktop	Mobile
Performance Score	~76	~49
Largest Contentful Paint	~7.3 sec	~8.1 sec
Total Blocking Time	~2.5 sec	~3.1 sec
Time to Interactive	~8 sec	~10 sec
Page Size	~4MB+	
Requests	~120+	

2. Image Optimization

Issues:

- - PNG/JPEG used — no WebP or AVIF
- - No lazy loading for off-screen images
- - Missing `width` and `height` attributes

- Several images >300 KB

Recommendations:

- - Convert all images to WebP or AVIF
- - Apply lossy compression (TinyPNG, Squoosh, ImageMagick)
- - Add `loading="lazy"` to all `<img>` elements
- - Use `<picture>` + `srcset` for responsive loading

- Define image dimensions to reduce layout shift

3. Technical & Code Optimizations

A. CSS & JS:

- - Minify and combine CSS and JavaScript
- - Remove unused CSS/JS with tools like PurgeCSS
- - Defer non-critical JS: `<script defer>`
- - Load Tailwind locally and build it using PostCSS

B. Fonts & Third-Party Scripts:

- - Host Google Fonts locally or use `display=swap`
- - Delay loading of non-essential widgets

C. HTML & DOM:

- - Reduce DOM size
- - Use semantic HTML tags
- - Avoid deeply nested divs

4. Server & Network Optimization

- - Enable Gzip or Brotli compression on server
- - Use a Content Delivery Network (CDN)
- - Enable browser caching for static assets

- Optimize backend queries for filters

5. Mobile & UX Improvements

- - Implement skeleton loaders or shimmer effects
- - Improve tap targets
- - Avoid horizontal scroll
- - Simplify mobile navigation menus

6. Error Handling & Debugging

- - Fix console errors (SVG, JS logs)
- - Remove unused preloads and dead scripts
- - Add fallback for filters and undefined data

- Set up 404 and error handling pages

7. Final Recommendations Summary

Area	Action Item	Priority
Images	Convert to WebP, lazy load, compress	High
Scripts	Minify, defer, remove unused JS	High
Styles	Minify CSS, use PurgeCSS	Medium
Fonts/Widgets	Load locally or defer	Medium
Server	Enable CDN + Brotli + cache static	High
Mobile Experience	Add skeletons, fix layout	High
Console Errors	Clean logs, fix SVG warnings	Medium
Filters/API	Fix undefined results	High

Conclusion

The Edu123 website offers valuable content but suffers from poor performance due to large image sizes, blocking scripts, and inefficient asset delivery. Implementing the above suggestions will drastically improve page speed, SEO, and user experience across devices.